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Can You Gig It? An Empirical Examination of the Gig Economy and Entrepreneurial Activity

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Abstract. We examine how the entry of gig-economy platforms influences local entrepreneurial activity. On the one hand, such platforms may reduce entrepreneurial activity by offering stable employment for the unemployed and underemployed. On the other hand, such platforms may enable entrepreneurial activity by offering work flexibility that allows the entrepreneur to redeploy resources strategically in order to pursue the nascent venture. To resolve this tension, we examine the entry of the ridesharing platform Uber X into local areas. We use two measures of entrepreneurial activity: crowdfunding campaign launches at Kickstarter, the world's largest reward-based crowdfunding platform, and levels of self-employment from the Current Population Survey. Results indicate a negative and significant relationship between platform entry and both measures of entrepreneurial activity. Importantly, the effect manifests primarily amongst unsuccessful Kickstarter campaigns and unincorporated entrepreneurial ventures, suggesting that gig-economy platforms predominantly reduce lower quality entrepreneurial activity, seemingly by offering viable employment for the unemployed and underemployed. These relationships are corroborated with a first-hand survey conducted with gig-economy service providers.

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Keywords: gig economy • digital platforms • crowdfunding • entrepreneurship • difference in difference • necessity-based entrepreneurship

Introduction

The introduction of new business models spawned by digital platforms has captured the attention of scholars and policy makers alike (Bakos and Bailey 1997, Parker and Van Alstyne 2005). And, while classic examples, such as eBay and Amazon.com (Brynjolfsson et al. 2003, Dellarocas and Wood 2008, Forman et al. 2008), continue to generate billions in revenue annually, new models of platform-enabled, peer-to-peer businesses have recently come to the fore (e.g., AirBNB, Postmates, Uber, TaskRabbit). Collectively referred to as the collaborative, sharing, or gig economy, businesses of this type are anticipated to comprise a substantial portion of the economy in the coming years with serious economic implications (Sundararajan 2014), including the disruption of long-standing industries (Morse 2015) and the displacement of incumbents (Zervas et al. 2015). Indeed, these platforms are already estimated to comprise a roughly \$26 billion market (Malhotra and Van Alstyne 2014), and numerous studies have explored patterns of demand for services provided by gig-economy platforms (Edelman and Luca 2014, Greenwood and Wattal 2017, Zervas et al. 2015). In this work, we extend this literature by offering, to our knowledge, the first consideration of how the entry of gig-economy platforms, formally defined as digital, on-demand platforms that enable a flexible

work arrangement,¹ influence local entrepreneurial activity.²

With industry disruption comes the expectation of economic growth, innovation, and entrepreneurship (Gans et al. 2002, McAfee and Brynjolfsson 2008). Yet little evidence of changes in entrepreneurial activity deriving from the gig economy have emerged (Zumbrun and Sussman 2015). While the absence of any apparent changes may be a result, at least in part, of the fact that traditional economic measures of employment and productivity are coarse and subject to lags, making it difficult to capture changes stemming from relatively recent developments (Sundararajan 2014, Badger 2013), it may also be a result of countervailing effects in the relationship between the gig economy and entrepreneurial activity.

To this point, both the popular press and the scholarly community provide competing arguments on the effect gig-economy platforms may have on entrepreneurial activity. On the one hand, the introduction of flexible ad hoc employment may lead to greater entrepreneurial activity because it affords the nascent entrepreneur the ability to strategically optimize the entrepreneur's time in order to garner the necessary resources to initiate a project or start a firm (Agrawal et al. 2015b, Douglas and Shepherd 2000). Both scholarly work and the popular press have repeatedly noted that gig-economy

platforms provide workers with an unprecedented degree of flexibility, allowing them to set their own schedules while earning stable pay (Hall and Krueger 2018, Johnson 2014, Tweney 2014). On the other hand, researchers have also noted that unemployment and underemployment are significant drivers of entrepreneurial activity. This is because people who do not have acceptable employment options may choose to engage in entrepreneurial activity because of their low opportunity costs (Acs and Armington 2006, Armington and Acs 2002, Fairlie 2002, Storey 1991). If this is the case, the arrival of the gig economy may reduce entrepreneurial activity by providing alternate employment opportunities for these entrepreneurs of necessity (Block and Koellinger 2009).

These two logics offer competing predictions of the effect gig-economy platform entry will have on local entrepreneurial activity, a tension we aim to resolve. More formally, we ask what is the effect of gig-economy platform introduction on the rate and characteristics of local entrepreneurial activity? To answer this question, we exploit the staggered entry of the ridesharing service Uber X into different markets to explore its relationship with two measures of entrepreneurial activity: (1) the volume of crowdfunding campaigns launched on Kickstarter, the world's largest crowdfunding platform (Burtch et al. 2013, 2015; Rhue 2015), and (2) self-employment reported in the U.S. Census Current Population Survey (CPS; Evans and Leighton 1989, Fairlie 2006) over a 36-month period between 2012 and 2015.

This econometric strategy offers us two notable benefits. First, because the rollout of Uber is staggered both temporally and geographically, that is, the platform enters different locations at different times, we can employ a difference-in-difference design that allows us to account for time-invariant, location-specific heterogeneity as well as broader macroeconomic trends that might otherwise influence our results. In doing so, we utilize the leads-and-lags model proposed by Autor (2003), which allows us to investigate whether significant pretreatment, market-level heterogeneity is biasing our estimations. Second, these two measures of entrepreneurial activity have unique data-generating processes. Thus, exploring both of them enhances the robustness and scope of conclusions that we can draw from the analysis. Further, each measure of entrepreneurial activity should be responsive to the introduction of gig-economy platforms as compared with other measures, such as firm founding and patenting (Bessen and Hunt 2007, Sundararajan 2014). Recall, CPS self-employment does not require the founding of a firm. Importantly, both measures enable an intuitive operationalization of entrepreneur quality. In the case of Kickstarter campaigns, we proxy for quality using the project's fundraising performance, relying on the market to separate high-quality projects from poor ones.

In the CPS, we proxy for quality based on whether the self-employed respondent operates an incorporated or unincorporated business as incorporation provides a strong signal of the formality of the organization.

Collectively, findings indicate that the entry of Uber X is associated with a significant decrease in both the volume of Kickstarter campaign launches and the level of self-employment. The relationship appears to manifest primarily among unsuccessful campaigns and unincorporated forms of self-employment. Intuitively, this suggests that some platform participants may be electing to work in the gig economy rather than pursue entrepreneurial projects of relatively low quality. The identified effects are pronounced. Locations where Uber X enters, on average, experience a 14% decline in the volume of campaigns launched on Kickstarter and a 5% decline in the level of self-employment over the year that follows. With regard to Kickstarter, this translates to a decrease of more than \$7.5 million in fundraising requests across the United States over the 36-month period of our study.³ With regard to the CPS, this implies a reduction of about 1,650 self-employed persons, relative to the sample median of 33,000, or a reduction of 4,350 self-employed persons relative to the sample mean of 87,000. We also supplement our archival analyses with first-hand data collected via a survey of 179 Uber and Lyft drivers. This survey allows us to assess the extent to which Uber X drivers have considered opening a small business as well as the extent to which Uber X drivers are aware of and utilize crowdfunding platforms, such as Kickstarter.

In addition to using the leads-and-lags model and direct survey of participants, we conduct a battery of robustness checks and falsification tests, including matching on observables, random implementation tests, the inclusion of city-specific linear time trends, and the estimation of hazard models predicting the entry of Uber. Results suggest not only that the entry of the platform is uncorrelated with pretreatment differences in local entrepreneurship, but that the negative relationship we observe is robust across a wide variety of empirical approaches.

This paper makes four notable contributions. First, the results speak to an important debate regarding how gig-economy platforms may influence local entrepreneurial activity. There are compelling reasons to believe that gig-economy platforms might either increase or decrease entrepreneurial activity. Although future work will be necessary to determine if all gig-economy platforms (e.g., AirBNB, Turo, Amazon Flex) exhibit similar effects, we provide initial evidence that gig-economy jobs may, on average, substitute for lower quality entrepreneurial activity rather than act as a complement to higher quality entrepreneurial activity. Second, we leverage a novel measure of entrepreneurial activity: the rate and volume of crowdfunding campaign launches (Mollick 2014). Unlike firm founding

or patenting, crowdfunding activity provides a more transparent, short-term bellwether of the rate and scale of entrepreneurship in a given geography. Further, we are able to demonstrate consistency between results based on this new measure and those based on a traditional measure of entrepreneurship (i.e., CPS self-employment). We therefore add to a nascent literature that makes a compelling case for the use of crowdfunding data as a measure of entrepreneurial activity (Mollick and Kuppaswamy 2014). Third, our work has implications for the sustainability of crowdfunding platforms, such as Kickstarter, on which sustainability and growth are tied to the crowd's efficient and successful identification of high-quality projects. Campaigns on crowdfunding platforms must compete for attention and capital. Given that gig-economy platform entry appears to reduce the number of lower quality crowdfunding campaigns, campaign backers' attention can refocus on the high-quality campaigns, thereby facilitating more efficient capital allocation (Shane 2009). Fourth, there are implications for policy. Findings suggest that gig-economy jobs may be particularly attractive to unemployed and underemployed individuals, who might otherwise have pursued low-quality entrepreneurial projects. Thus, although the introduction of new digital platforms may speed the demise of incumbents (Seamans and Zhu 2013), possibly eliminating some jobs in the process, our results suggest that individuals with weak attachments to the labor market may benefit from this creative destruction. This yields important insights for policy makers as they tackle the regulation and legality of these platforms.

Our analysis depends on key assumptions, the most notable being (i) that, conditional on controls, the entry and timing of Uber X is exogenous with respect to our measures of entrepreneurial activity, and (ii) individuals who drive for Uber X would otherwise have engaged in self-employment or would have sought funding on Kickstarter. The endogeneity of Uber's entry is thus a central concern. Accordingly, establishing the robustness of our results to a variety of empirical specifications and measures, in addition to corroborating our archival findings with primary data from drivers themselves, is critical. Collectively, we present a portfolio of consistent evidence, but as with any archival study, our results should be interpreted with caution. Nevertheless, we believe the constellation of evidence we present provides a compelling and consistent picture of the relationship between the gig economy and entrepreneurial activity. Moreover, this research serves as an important initial step in furthering our understanding about the relationship between the gig economy and local economic activity, one that can be built upon by future researchers. We hope this research serves as a call for future scholars to dive

deeper into the supply side of the gig economy and the relationship between gig-economy platforms and broader macroeconomic outcomes.

Related Work

Academic research on the gig economy has proceeded along a number of fronts, ranging from platform design and user response (Fradkin 2015, Fradkin et al. 2015) to the effect on labor movements (Friedman 2014, Milkman and Ott 2014) to broader economic and societal effects (Edelman and Luca 2014, Edelman et al. 2017, Greenwood and Wattal 2017, Zervas et al. 2015). The latter effects, in particular, have received considerable attention. Zervas et al. (2015), for example, examine the impact of AirBNB's entry on the Texas hotel industry, finding strong evidence of cannibalization, particularly among lower-tier hotels. Cramer and Krueger (2016) explore the drivers of Uber's and Lyft's similar cannibalization of the traditional taxi-cab industry. Greenwood and Wattal (2017) study the effect of Uber entry on DUI fatalities and report similar evidence: that effects primarily accrue at the bottom of the demand curve among the most price-sensitive consumers. Edelman et al. (2017) further report evidence of racial discrimination on AirBNB, finding that guests with distinctively African-American names are less likely to be accepted for lodging.

However, and strikingly, there is a conspicuous shortage of work that considers the supply side of these markets. In particular, researchers rarely consider who the suppliers of these services are likely to be with three notable exceptions. Edelman and Luca (2014) examine how racial bias affects AirBNB hosts, Rhue (2015) and Younkin and Kuppaswamy (2018) examine racial bias in crowdfunding, and Morse (2015) discusses the impacts that P2P lending markets have on consumer lending.

In this work, we consider how gig-economy platforms may influence entrepreneurial activity. Although research, thus far, has focused on the economic benefits that the gig economy may produce, in terms of flexible employment and micro-entrepreneurship (Sundararajan 2014), it is still unclear why individuals choose to supply labor on these platforms. Relatedly, it is also unclear what activities these individuals might have engaged in had the platform not been available. Determining the answers to these questions is critical to understanding the macro-level trade-offs that accompany the operation of gig-economy platforms. Further, understanding the nature of supplier characteristics may help researchers and policy makers to better understand how these markets operate.

Slack Resources, Experimentation, and Entrepreneurship

Why might the entrance of gig-economy platforms increase the amount of local entrepreneurial activity?

The prior literature offers two mechanisms. First, scholars have argued that entrepreneurship depends on the availability of slack resources, that is, resources that can be directly reassigned to entrepreneurial endeavors (Agrawal et al. 2015b, Richtnér et al. 2014). Because gig-economy platforms purportedly offer service providers a combination of scheduling flexibility and stable income, their entry may enable the would-be entrepreneur to strategically reallocate a constrained sets of resources, for example, time, to push a nascent idea forward (Agrawal et al. 2015b). Second, and related to the first point, the nascent entrepreneur, unburdened by earlier resource constraints, may be encouraged to explore or exploit new opportunities as they emerge (Shah and Tripsas 2007, Voss et al. 2008). In principle, gig-economy employment may therefore enable experimental sampling of potential opportunities because sufficient resources are available to afford it (Greve 2007, Kerr et al. 2014, Shah and Tripsas 2007).

Significant evidence of both behaviors can be found in the empirical literature and the popular press. Agrawal et al. (2015b), for example, find that when prestigious universities release their students for breaks, for example, winter or summer vacations, there is a dramatic uptick in the amount of entrepreneurial activity around those universities—a notion that is corroborated by Facebook and Microsoft both being founded when Harvard was in winter session (Graham 2012). Similarly, Voss et al. (2008) find, in the context of professional theatre companies, that slack resources tend to be redeployed to explore nascent opportunities. On the practitioner side, anecdotal evidence underscores the success of Google's, Hewlett-Packard's, and 3M's "free time policies," which have led to a number of valuable products.

In the context of the gig economy, this logic is compelling and suggests that the flexibility of gig-economy employment would allow the nascent entrepreneur to optimize the entrepreneur's resources in such a way that the entrepreneur could redeploy time and other assets to a budding venture (Douglas and Shepherd 2000). Put another way, because gig-economy firms such as Uber allow the entrepreneur to set the entrepreneur's own hours (Hall and Krueger 2018), they offer a significant advantage over traditional employers with less-flexible schedules (Swarns 2014). The entrepreneur may be able to exploit this flexibility for the entrepreneur's own gain and devote resources to the venture without losing financial security.

Opportunity Costs and Entrepreneurship

Although the literature on slack resources would suggest that the entrance of gig-economy platforms might facilitate entrepreneurial entry, other work provides the opposite prediction. While many would-be entrepreneurs might depart traditional employment to

pursue higher wages (Braguinsky et al. 2012, Carnahan 2013) or increased flexibility (DeMartino and Barbato 2003, Sørensen 2007), past work also suggests that individuals may pursue entrepreneurship as a means of resolving unemployment or underemployment (Block and Koellinger 2009, Fairlie 2002, Storey 1991).

Received research has argued that unemployed and underemployed individuals pursue entrepreneurial activity because they have significantly lower opportunity costs than someone who is fully employed (Block and Koellinger 2009, Fairlie 2002, Storey 1991). As a result, they may pursue entrepreneurial activity because they have excess time (as a result of partial employment) or because such activity has a higher expected value than their wage employment opportunities (Acs and Armington 2006, Armington and Acs 2002). The entrance of employment opportunities via the gig economy may change the internal calculus for these would-be entrepreneurs. As the gig economy may offer significant employment opportunities (Uber employed roughly 327,000 drivers in the United States as of the end of 2015 (Carson 2015) and 7.9% of U.S. workers worked in contingent employment as of 2010 (GAO 2015)), it is plausible that entrepreneurial activity may fall as a result of the gig economy because entrepreneurs who are unemployed or underemployed select into this new, viable alternative. Put another way, if individuals with lower opportunity costs pursue employment in the gig economy rather than pursuing entrepreneurial activity, then it is plausible that total local entrepreneurial activity may *fall* as the gig economy grows.

The presence of persuasive theoretical arguments on both sides of this debate reveals a compelling tension. On the one hand, entrepreneurial activity may rise because individuals can use the slack resources that are freed up via employment in the gig economy to pursue new projects. On the other, if the gig economy predominantly provides economic opportunities to the unemployed or underemployed, entrepreneurial activity may decline as those individuals redeploy their efforts away from entrepreneurial activity toward gig-economy employment. Therefore, in the absence of a compelling *a priori* expectation, we look to our empirical analyses to determine the predominant effect.

Methods

Data and Descriptive Statistics

We draw on data from multiple sources to execute our estimations. To construct our first measure of entrepreneurial activity, we capture data on the population of crowdfunding campaigns launched on Kickstarter. The campaign data are collected via Kickstarter's API⁴ through daily queries of active projects from January 2012 through December of 2015. By paging through the active campaign list, we capture every campaign that

was live at any given point in time. In turn, we identify the date, location, and amount of money sought and ultimately raised by each campaign. To construct our second measure of entrepreneurial activity, we leverage the count of self-employed persons in each location using the public-use version of the Current Population Survey (CPS).⁵ These data have been used extensively in prior empirical research (Fairlie 2013, Lazear 2004). We include all occupations in our construction of this measure although the results are unchanged if we exclude driving-related occupations (who might be Uber drivers⁶). To construct control variables, we obtain dynamic socioeconomic data across locations from the CPS and the Area Resource File.⁷

We use Kickstarter campaign launches and census self-employment to measure entrepreneurial activity for several reasons. First, each measure has been used in the existing literature (Mollick 2014, Parker 2004) although Kickstarter less commonly so. Second, these measures are highly complementary. Kickstarter measures entrepreneurial activity at the project level and is capable of capturing entrepreneurial activity for individuals who are otherwise engaged in wage employment. CPS self-employment captures entrepreneurial activity that is significant enough to be an individual's primary work. In addition, Kickstarter captures entrepreneurial activity that is perhaps less traditional than that measured by the CPS.⁸ The data-generating process for each of these measures is also distinct, thereby increasing the robustness of the empirical test. Kickstarter is based on the posting of a project on the platform whereas the CPS is based on monthly household surveys.

Further, each measure captures a wide spectrum of entrepreneurial activity in terms of quality. This variation allows us insight into whether individuals may be engaging in opportunity or necessity entrepreneurship. Alternative measures, such as VC financing or patent production, exclusively capture successful attempts as obtaining VC financing and/or a patent is rare (Robb and Robinson 2014). Kickstarter and CPS should also be responsive to changes in local market conditions, as it is relatively simple to launch a Kickstarter campaign, and individuals can select into self-employment with relative ease. Finally, the timing of each of these types of activity can be measured precisely because one can observe the exact day when a Kickstarter campaign is posted, and the CPS provides monthly data files.

Our key independent variable of interest is the entrance of the ridesharing service, Uber X, into a local area. Uber, founded in 2009, is a smartphone application that allows consumers to submit a trip request, which is then routed to Uber drivers who use their own cars to fulfill the request. Uber X is Uber's low-cost option. Data on Uber entry are retrieved directly from

the Uber blog. We focus on Uber X for several reasons. First, Uber rolled out on a city-by-city basis; thus, the expansion of the platform is geographically and temporally dispersed. In contrast, many other platforms, such as AirBNB, were rolled out without geographic restriction. This prevents the identification of a plausible control group, without which it would be difficult to estimate the effect of platform entry. Second, Uber is the largest platform in the gig economy with a valuation of nearly \$70B as of early 2016. Large platforms are preferable for our analyses because they are more likely to impact local labor dynamics.⁹ Third, unlike many of the other gig-economy platforms that have engaged in staggered rollouts, Uber is particularly systematic about publishing the dates and locations of rollouts, which has the benefit of clear measurement. Finally, and perhaps most importantly, Uber X purposefully enables a flexible work arrangement, thereby meeting our formal definition of the gig economy. This stands in contrast to platforms such as Etsy, which would more appropriately be classified as online retailers.

Econometric Specification

It is unlikely that Uber X randomly enters cities, and features of a location (e.g., size, location, economic characteristics) may make it an attractive target for Uber's entry while simultaneously affecting local entrepreneurial activity. This could, in turn, result in selection effects and bias in our estimations. Accordingly, we take several steps to safeguard against the possible bias induced by selection in our analyses. First, we include location-specific fixed effects, which control for any time-invariant omitted factors that characterize a city, and time-fixed effects, which control for United States-wide unobserved temporal trends or shocks (e.g., seasonality). Second, we control for time-varying characteristics of the local economy, such as size, wages, unemployment, and education levels. Finally, and perhaps most importantly, the primary econometric specification we employ is a multi-site entry, difference-in-difference (DD), relative time model (Angrist and Pischke 2008, Autor 2003). This leads-and-lags model allows us to conduct a quasi-experiment using secondary data because the treatment, that is, the entry of Uber X, is applied in different locations at different times. This allows us to use untreated locations, that is, locations that Uber has not entered, as controls for treated locations, that is, locations with Uber. By exploiting the longitudinal nature of the data and examining pretreatment trends in entrepreneurial activity, we can explicitly test the validity of untreated locations we employ as the control (Bertrand et al. 2004).

Our unit of analysis is the economic area (EA)-month. The Bureau of Economic Analysis (1995) divides the United States into 172 unique economic areas

based on shared economic activity (in particular, commuting patterns). Our first dependent variable (DV) is the number of Kickstarter campaigns launched in EA i during month t . Our second dependent variable is the number of self-employed individuals in EA i during month t . The results also include survey weights, which the CPS provides to make the data representative of the overall U.S. population although unweighted results are consistent.

In our initial analyses, the independent variable of interest is the treatment indicator, *Uber X*, which indicates that the ridesharing service *Uber X* has entered EA i as of time t . Consistent with prior work examining the effect of Uber's entry on a locale (Greenwood and Wattal 2017), we focus on *Uber X* rather than the premium service *Uber Black* because of the significantly lower startup costs and larger network of drivers.¹⁰ *Uber Black* requires a driver to use a "black car" while an *Uber X* driver can utilize a broader range

of lower cost, personal vehicles. A full list of Uber entries by location is available in Table 1. To complete the difference-in-difference specification, we include vectors of EA and time-fixed effects as well as time-varying controls for the characteristics of the local area.

Our primary sample covers 2012–2015, during which 123,569 Kickstarter campaigns were launched in the United States. We start the analysis in 2012 because this is the first year in which Kickstarter experiences significant campaign volume, and we end the analysis in 2015 because this is the last year for which CPS data files are available at the time we conducted the analysis. While there are 172 EAs in the United States, our primary sample covers 144 of the largest EAs. The remaining 28 EAs are omitted because the public use CPS data does not publish information on smaller EAs (because identities of individuals might be discernable). Table 2 provides the descriptive statistics for our sample. Figure 1 is a histogram of *Uber X* entries.¹¹

Table 1. EAs Experiencing Uber X Entry

EA	Uber X	EA	Uber X
Boston, MA	6/1/2013	St. Louis, MO	10/9/2014
New York, NY	8/1/2013	Kansas City, MO	5/1/2014
Philadelphia, PA	10/15/2015	Des Moines, IA	9/1/2014
Washington, DC	8/1/2013	Madison, WI	3/1/2014
Richmond, VA	8/1/2014	Minneapolis, MN	9/1/2013
Greensboro, NC	8/1/2014	Omaha, NE	5/1/2014
Raleigh-Durham, NC	6/1/2014	North Platte, NE	8/1/2014
Norfolk, VA	7/1/2014	Wichita, KS	8/1/2014
Charlotte, NC	9/1/2013	Tulsa, OK	3/1/2014
Charleston, SC	6/1/2014	Oklahoma City, OK	9/1/2013
Jacksonville, FL	5/1/2014	Dallas-Fort Worth, TX	11/1/2013
Orlando, FL	6/1/2014	Austin, TX	8/1/2014
Miami, FL	6/1/2014	Houston, TX	2/1/2014
Tampa, FL	4/1/2014	Corpus Christi, TX	6/1/2014
Atlanta, GA	10/1/2013	San Antonio, TX	3/1/2014
Knoxville, TN	8/1/2014	Lubbock, TX	6/1/2014
Lexington, KY	6/1/2014	Denver, CO	10/1/2013
Cincinnati, OH	3/1/2014	Spokane, WA	5/1/2014
Dayton OH	8/1/2014	Boise, ID	10/1/2014
Columbus, OH	2/1/2014	Salt Lake City, UT	5/1/2014
Pittsburgh, PA	2/1/2014	Flagstaff, AZ	9/1/2014
Cleveland OH	4/1/2014	Albuquerque, NM	5/28/2014
Toledo, OH	6/1/2014	El Paso, TX	6/1/2014
Detroit, MI	10/1/2013	Phoenix, AZ	8/1/2013
Grand Rapids, MI	7/1/2014	Tucson, AZ	2/1/2014
Milwaukee, WI	3/1/2014	Los Angeles, CA	9/1/2013
Chicago, IL	4/1/2013	San Diego, CA	5/1/2013
Fort Wayne, IN	8/1/2014	Fresno, CA	2/1/2014
Indianapolis, IN	9/1/2013	San Francisco, CA	7/1/2012
Louisville, KY	4/1/2014	Sacramento, CA	11/1/2013
Nashville, TN	12/1/2013	Eugene, OR	7/1/2014
Memphis, TN	4/1/2014	Portland-Salem, OR	7/1/2014
Birmingham, AL	8/1/2014	Seattle, WA	4/1/2013
Montgomery, AL	8/28/2014	Anchorage, AK	9/16/2014
New Orleans, LA	4/16/2015	Honolulu, HI	6/1/2014
Baton Rouge, LA	7/1/2014		

Note. EA names are shortened to the largest city in the EA in the interest of space.

Table 2. Summary Statistics

Variable	Obs.	Mean	Std. dev.	1	2	3	4	5	7	8	9	10	11
1 <i>Campaigns</i>	8,304	15.71	44.54	1.00									
2 <i>Self-employment</i>	6,731	87,792	167,235	0.82	1.00								
3 <i>Taxi employment</i>	6,731	2,618	9,317	0.72	0.78	1.00							
4 <i>Dollars pledged</i>	8,304	178,207	787,864	0.74	0.69	0.56	1.00						
5 <i>Uber X</i>	8,304	0.17	0.38	0.42	0.31	0.22	0.25	1.00					
6 <i>Employment</i>	6,731	858,549	1,539,087	0.81	0.98	0.83	0.66	0.32	1.00				
7 <i>Avg. weekly wage</i>	8,304	715.86	94.16	0.41	0.46	0.39	0.36	0.31	0.47	1.00			
8 <i>Total quarterly wages (\$M)</i>	8,304	7,311.18	15,259.83	0.63	0.83	0.61	0.54	0.20	0.83	0.42	1.00		
9 <i>Unemployment rate</i>	6,731	0.06	0.03	−0.02	0.06	0.03	0.02	−0.14	0.05	−0.04	0.09	1.00	
10 <i>% with bach. degree</i>	6,731	0.43	0.12	0.21	0.22	0.17	0.16	0.21	0.22	0.17	0.20	−0.21	1.00

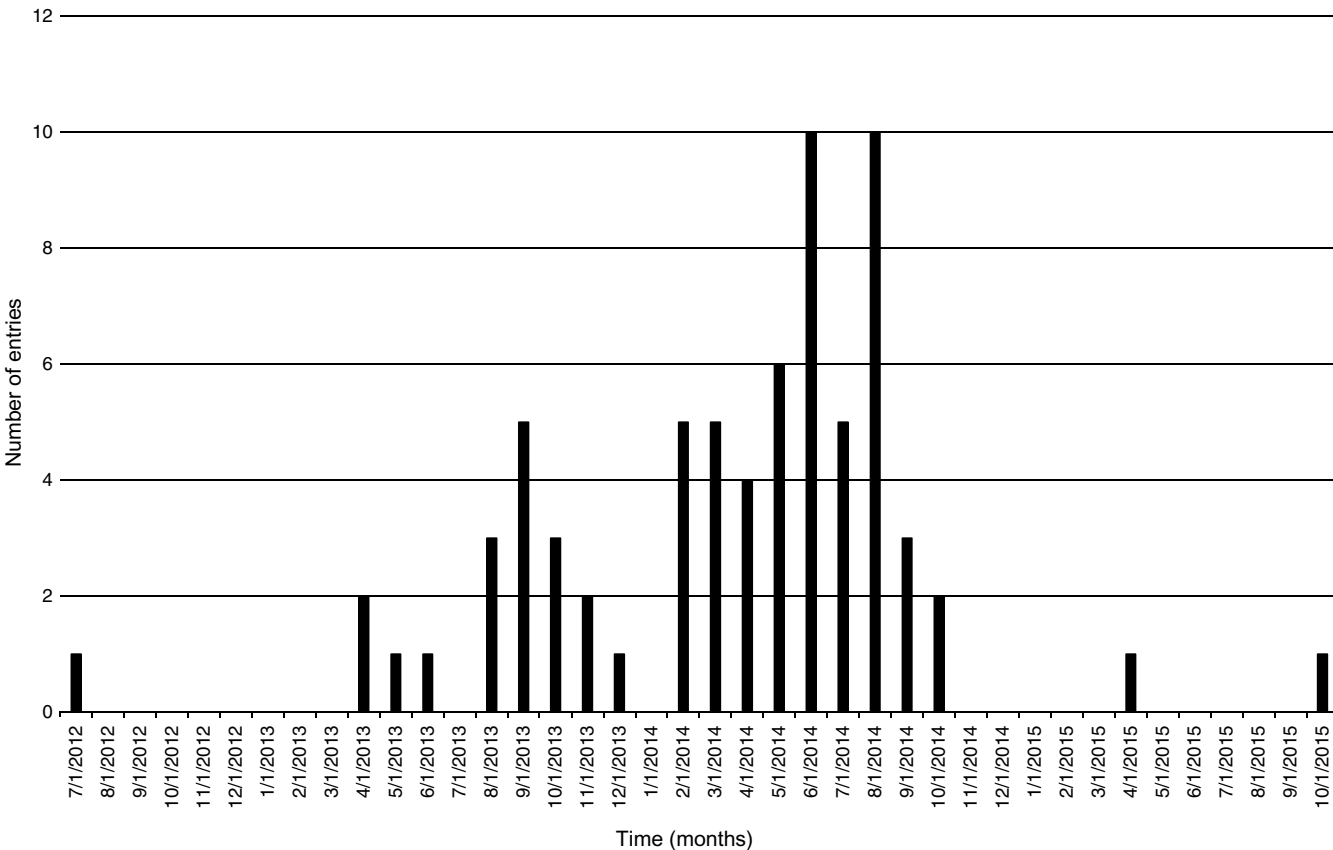
Note. Level of analysis is EA-month.

Our estimation incorporates a second set of time dummies that indicate the chronological distance, j , of an observation period, t , with respect to the timing of treatment in that location, i . Our model specification is expressed in linear form for the sake of simplicity in Equation (1). Here, y_{it} is the number of campaigns launched in EA i at time t or the number of self-employed persons in EA i at time t ; α_i represents the vector of EA fixed effects, ϕ is the vector of relative time dummies, τ is a vector of unique quarter dummies, and X is a series of dynamic location-specific

controls. For the relative time dummies, to tabulate all of the effects, we collapse all pretreatment periods that are greater than or equal to seven quarters prior to treatment into one dummy and all posttreatment periods that are greater than or equal to seven quarters following treatment into another dummy. In the seven-quarter pretreatment dummy, there are 69 cities, and in the seven-quarter posttreatment dummy, there are 24 cities.

$$y_{it} = \alpha_i + \tau_t + \sum_j \beta_j(\phi_{it}) + X_{it} + \varepsilon_{it}. \quad (1)$$

Figure 1. Histogram Uber X Platform Entries



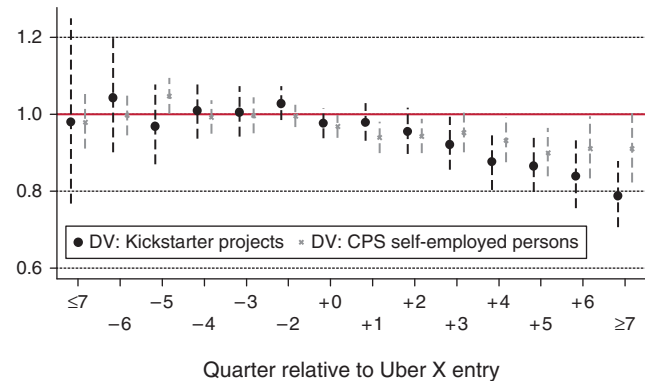
As noted previously, it is unlikely that Uber randomly selects markets for entry. While an evaluation of the parallel trends assumption helps us to assess the validity of our identification strategy, it remains possible that Uber selects into locations based on dynamic factors that might also correlate with entrepreneurial activity. To account for this possibility, we also incorporate a number of location-specific dynamic control variables (X), which capture factors that might make a market attractive to Uber while also influencing the rate of local entrepreneurial activity. These controls include the average weekly wage within the EA, total quarterly wages within the EA, the unemployment rate in the EA, the percentage of EA residents who have at least a bachelor's degree, and the number of employed people working in the EA. The first two controls come from the Area Resource File while the latter three come from the CPS.

Our primary estimator is a fixed-effects Poisson pseudo-maximum likelihood (or quasi-maximum likelihood) estimator (PPML or PQML) (Azoulay et al. 2010, Burtch et al. 2014, Greenwood and Gopal 2015, Simcoe 2007). Although a negative binomial (NB) estimator is typically prescribed in cases of over-dispersed count data, as discussed by Allison and Waterman (2002), no true fixed-effect estimator has yet been proposed in the NB case whereas a conditional fixed-effect Poisson estimator is available. Moreover, the PPML estimator enables us to obtain consistent, robust standard errors (clustered by EA) even under conditions of over-dispersion (Wooldridge 1997). The PPML estimator has also been shown to significantly outperform a log-OLS specification when data contains many zeroes (Silva and Tenreiro 2011), as in our sample, and to provide more reliable estimates. This is because a log-OLS specification in count data can produce severely biased estimates (O'Hara and Kotze 2010), particularly under conditions of heteroscedasticity (Silva and Tenreiro 2006).

Results

Results in Table 3 reveal several interesting findings. First, none of the models using Kickstarter campaign volumes as a dependent variable exhibit a statistically significant pretreatment trend, and only one pretreatment relative time dummy in the CPS self-employment models exhibits statistical significance (this is eliminated when we apply a matching procedure). Second, results are insensitive to sample definition and the inclusion of control variables. The primary sample covers the 144 largest EAs from 2012 to 2015 for reasons previously described. Model 1 extends the sample geographically to cover all 172 EAs (employing the Kickstarter measure) while Model 4 extends the sample temporally to cover 2010 through 2015 (using the CPS measure). Models 1, 2, 4, and 5 do not contain control variables while Models 3 and 4 do. Results

Figure 2. (Color online) Effect of Uber X on Kickstarter Campaign Launches and CPS Self-Employment



Notes. 95% confidence intervals. Results come from Table 3, Models 3 and 6. Poisson regressions with EA fixed effects, quarter dummies. Controls include number of employed persons, average wages, total quarterly wages, unemployment rate, and percentage of persons with at least a bachelor's degree. Standard errors clustered by EA.

are consistent across all specifications. We see that the negative effect of *Uber X* entry on the number of Kickstarter campaigns becomes stable and significant roughly one year after implementation, consistent with previous estimates of the effect of Uber (Greenwood and Wattal 2017). Specifically, estimates in Model 1 suggest that the number of active Kickstarter projects declines roughly 14% in treated EAs in the fourth quarter after *Uber X*'s entry while estimates in Model 4 suggest that the number of persons reporting self-employment in the CPS declines roughly 5%. Graphical representations of the estimates are depicted in Figure 2. Broadly speaking, these estimates provide strong and significant evidence that the entry of gig-economy platforms negatively associates with entrepreneurial activity in local areas.¹²

Additional Analyses and Robustness Tests

We next consider an extensive set of robustness checks, which are summarized in Table 4. We begin by matching treated EAs to untreated EAs in order to ensure that our treatment and control groups are comparable. We then address the possibility that our results are spurious, deriving from serial correlation in the dependent variable (Bertrand et al. 2004) via a random implementation test. Third, we explore the robustness of our results to controlling for entry by competing ridesharing services, such as Lyft and Curb, as well as the effect of Uber Black. Fourth, we consider additional controls in our relative time specification, namely location-specific time trends.

Following the baseline analyses, we explore a set of supplemental, secondary analyses intended to shed light on the mechanisms underlying the relationship.

Table 3. Relative Time Model of the Effect of Uber X Entry on Kickstarter Campaign Launches and CPS Self-Employment Levels

DV:	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	<i>Campaigns</i>	<i>Campaigns</i>	<i>Campaigns</i>	<i>Self-Employed Persons</i>	<i>Self-Employed Persons</i>	<i>Self-Employed Persons</i>
7+ quarters until Uber X entry	−0.0217 (0.128)	−0.0246 (0.131)	−0.0206 (0.124)	−0.00664 (0.0269)	−0.0128 (0.0385)	−0.0207 (0.0373)
6 quarters until Uber X enters	0.0394 (0.0757)	0.0350 (0.0780)	0.0420 (0.0745)	0.00684 (0.0234)	0.000881 (0.0276)	−0.00269 (0.0276)
5 quarters until Uber X enters	−0.0290 (0.0548)	−0.0340 (0.0565)	−0.0320 (0.0546)	0.0521* (0.0211)	0.0469* (0.0233)	0.0459* (0.0229)
4 quarters until Uber X enters	0.00843 (0.0391)	0.00478 (0.0402)	0.0102 (0.0389)	7.90e−05 (0.0223)	−0.00582 (0.0241)	−0.00813 (0.0224)
3 quarters until Uber X enters	−0.000369 (0.0335)	−0.00296 (0.0343)	0.00572 (0.0332)	0.00259 (0.0235)	0.00160 (0.0253)	−0.00369 (0.0241)
2 quarters until Uber X enters	0.0197 (0.0201)	0.0224 (0.0201)	0.0277 (0.0219)	−0.000360 (0.0153)	−0.00109 (0.0156)	−0.00443 (0.0149)
1 quarter until Uber X enters				Omitted		
0 quarters since Uber X entry	−0.0296 (0.0218)	−0.0280 (0.0220)	−0.0236 (0.0205)	−0.0309 (0.0158)	−0.0305 (0.0159)	−0.0322* (0.0157)
1 quarters since Uber X entry	−0.0342 (0.0332)	−0.0306 (0.0337)	−0.0212 (0.0255)	−0.0549* (0.0225)	−0.0546* (0.0221)	−0.0631** (0.0223)
2 quarters since Uber X entry	−0.0567 (0.0385)	−0.0538 (0.0393)	−0.0460 (0.0322)	−0.0551* (0.0227)	−0.0528* (0.0240)	−0.0592* (0.0243)
3 quarters since Uber X entry	−0.0886 (0.0473)	−0.0854 (0.0480)	−0.0813* (0.0380)	−0.0455 (0.0253)	−0.0419 (0.0276)	−0.0496 (0.0280)
4 quarters since Uber X entry	−0.141** (0.0516)	−0.137** (0.0526)	−0.131** (0.0446)	−0.0621* (0.0281)	−0.0574 (0.0322)	−0.0703* (0.0319)
5 quarters since Uber X entry	−0.161** (0.0554)	−0.157** (0.0569)	−0.145** (0.0416)	−0.0941** (0.0310)	−0.0875* (0.0353)	−0.106** (0.0358)
6 quarters since Uber X entry	−0.193** (0.0630)	−0.190** (0.0649)	−0.175** (0.0535)	−0.100** (0.0389)	−0.0876 (0.0467)	−0.0942* (0.0452)
7+ quarters since Uber X entry	−0.260*** (0.0504)	−0.257*** (0.0529)	−0.239*** (0.0558)	−0.142** (0.0449)	−0.124* (0.0536)	−0.0937 (0.0518)
Employment controls from Table 2	No	No	Yes	No	No	Yes
Observations	8,256	6,731	6,731	10,115	6,731	6,731
Number of EAnum	172	144	144	144	144	144
Model	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
EA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time period	2012–2015	2012–2015	2012–2015	2010–2015	2012–2015	2012–2015

Notes. Dependent variable *Campaigns* indicates the DV is the number of Kickstarter campaigns launched in *it*. Dependent variable *Self-Employed Persons* is the number of self-employed individuals in *it*. Employment controls indicate the number of employed people, average weekly wage, total quarterly wages within the EA, percentage of unemployed people in the EA, and percentage of individuals in the EA with at least a bachelor's degree. Estimator is a PPML. Standard errors, in parentheses, are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

In doing so, we consider heterogeneity with respect to the quality of the Kickstarter campaigns and CPS-reported self-employment. Additionally, we explore changes in driver employment as reported in the CPS.

Matching Treated and Untreated EAs

One issue with the prior estimations is that they rely on an assumed functional form in the relationship between our control variables and our outcome measures. Violation of these assumptions can lead to model dependence and thus bias our estimates. To help

address this potential problem, we preprocess our data by matching treated EAs to untreated EAs. Specifically, we match each EA that receives Uber X to an EA that never receives Uber X based on the volume of Kickstarter campaigns or individuals reporting self-employment on the CPS (depending on the outcome measure to be estimated) in the two quarters prior to Uber X's entry. We then enforce this match for the city's entire estimation panel.¹³ If a control EA is matched to multiple treated EAs, we retain the pair in which the treatment occurs earliest.

Table 4. Summary of Robustness Tests

Concern	Test	Finding	Location
Serial correlation in standard errors	Random implementation (shuffle) test	Placebo treatments yield no correlation with campaign volume	Table 6
Cities receiving treatment experiencing different economic trajectory	Replicate estimations with EA-specific time trends	Results remain consistent	Table 9
	Shuffle Test swapping across treated cities	No significant correlation with campaign volume	Table 7
	Examine predictors of Uber X entry into an EA	Levels and changes in entrepreneurial activity not related to Uber X entry	Table 12
Outliers driving the results (large cities)	Replicate estimations without New York, Boston, and San Francisco	Results remain consistent	Available upon request
Lack of comparability between treated and control groups	Match cities that receive Uber X to cities that do not	Results remain consistent	Table 5
Alternate ridesharing services driving effect	Replicate estimations including other discount ridesharing services	Results remain consistent	Table 8
Uber X not representative of gig economy	Replicate estimations using courier service Postmates	Results remain consistent in CPS, results exhibit pretreatment trends with Kickstarter	Available upon request
Reduction in campaigns not a function of necessity entrepreneurship	Examine heterogeneity of effect across entrepreneurs of varying quality	Results primarily accrue to lower quality campaigns (Kickstarter) and unincorporated self-employment (CPS)	Table 10
	Replicate with higher startup cost platform (Uber Black)	No significant correlation with campaign volume	Table 8
Inability to track individual drivers and entrepreneurs	Small-scale survey of current Uber drivers	Nontrivial portions of drivers indicate that they have considered launching a venture	See final subsection of empirical section
	Replicate estimates with drivers and chauffeurs in local area as DV	Number of local people claiming to drive professionally increases	Table 11
Incorrect functional form of time controls	Replicate with month and month-year time controls	Results remain consistent	Available upon request
Functional form of model is not Poisson	Replicate using logged OLS	Results consistent in CPS	Available upon request

Results are in Table 5. First, we note that none of the pretreatment periods is statistically significant once we apply the matching procedure. Second, we note that results otherwise remain qualitatively similar in terms of pattern and effect size compared with what we observe in the unmatched sample. For example, it takes about four quarters for the effect to appear, and we see an 8% reduction in Kickstarter campaigns (Model 2) as well as a 10% reduction in CPS self-employment (Model 4) one year following the entry of Uber X. Models 1 and 3 report difference-in-difference estimates, which collapse the posttreatment period into a single dummy with similar results.

Random Implementation (Shuffle) Tests

Next, we consider the potential for false significance in our estimates as a result of spurious relationships or serial correlation in our dependent variable. As discussed by Bertrand et al. (2004), “Most papers that employ Differences-in-Differences estimation (DD) use many years of data and focus on serially correlated outcomes but ignore that the resulting standard errors are

inconsistent” (p. 249). Because a large portion of the market entries observed in our data are concentrated in early 2014 (Figure 1), it is possible that something idiosyncratic about these cities is driving the observed effect (such as a particular reaction to the 2014 economic recovery from the Great Recession). Although our initial estimates (Table 3) cluster standard errors by EA and thus account for these dependencies to some degree, it is useful to implement several of the tests suggested by the Bertrand et al. (2004)¹⁴ to ensure robustness.

In the first analysis, a random implementation test, we reassign our treatment indicators at random in our data. That is, we shuffle the 1,440 indicators of Uber X’s presence in our original sample to a randomly selected set of (new) observations, thereby creating a pseudo (placebo) treatment. We then estimate a standard difference-in-difference model (i.e., an Uber presence dummy) with time and location fixed effects. We store the coefficient of this pseudo-treatment and replicate the procedure 1,000 times. This test helps

Table 5. Coarsened Exact Match Relative Time Model of the Effect of Uber X Entry on Kickstarter Campaign Launches and Self-Employment Levels

DV:	Model 1 <i>Campaigns</i>	Model 2 <i>Campaigns</i>	Model 3 <i>Self-Employed Persons</i>	Model 4 <i>Self-Employed Persons</i>
Post Uber X entry	0.146** (0.0523)		0.0655 (0.0546)	
Post Uber X * Treated	−0.118* (0.0568)		−0.114* (0.0554)	
≥7 quarters until Uber X entry		−0.0821 (0.129)		−0.0656 (0.136)
6 quarters until Uber X enters		−0.00193 (0.0799)		0.0113 (0.0834)
5 quarters until Uber X enters		−0.0669 (0.0572)		−0.0551 (0.0591)
4 quarters until Uber X enters		−0.0158 (0.0410)		−0.00840 (0.0436)
3 quarters until Uber X enters		−0.0111 (0.0346)		−0.00477 (0.0359)
2 quarters until Uber X enters		0.0208 (0.0227)		0.0230 (0.0233)
1 quarter until Uber X enters		Omitted		Omitted
0 quarters since Uber X entry		−0.00840 (0.0214)		−0.0151 (0.0214)
1 quarters since Uber X entry		−0.00195 (0.0266)		−0.00663 (0.0274)
2 quarters since Uber X entry		−0.0155 (0.0343)		−0.0278 (0.0357)
3 quarters since Uber X entry		−0.0429 (0.0402)		−0.0582 (0.0429)
4 quarters since Uber X entry		−0.0840+ (0.0483)		−0.106* (0.0512)
5 quarters since Uber X entry		−0.0872+ (0.0471)		−0.115* (0.0509)
6 quarters since Uber X entry		−0.112+ (0.0588)		−0.143* (0.0625)
≥7 quarters since Uber X entry		−0.161* (0.0635)		−0.197** (0.0679)
Employment controls from Table 2	Yes	Yes	Yes	Yes
Observations	4,702	4,702	4,306	4,306
Number of EAnum	99	99	91	91
Model	Poisson	Poisson	Poisson	Poisson
EA fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes
Time period	2012–2015	2012–2015	2012–2015	2012–2015

Notes. Dependent variable *Campaigns* indicates the DV is the number of Kickstarter campaigns launched in it . Dependent variable *Self-Employed Persons* is the number self-employed individuals in it . Employment controls indicate the number of employed people, average weekly wage, total quarterly wages within the EA, percentage of unemployed people in the EA, and percentage of individuals in the EA with at least a bachelor's degree. Treated and control EAs are matched in the quarter prior to Uber X entry. Matching is based on the level of the DV in the two quarters prior to Uber X entry, and control EAs are restricted to those that never receive Uber X during the sample period. Estimator is a PPML. Standard errors in parentheses are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.10$.

to assess the spuriousness of any significant results obtained in the main analyses arising from autocorrelation in the dependent variable (Bertrand et al. 2004).

Results are presented in Table 6 and indicate three key findings. First, the estimated average β associated with the randomly assigned treatment indicator is not

significantly different from zero, suggesting that the main results are reliably estimated. Second, the estimated β is quite small and not driven by serial correlation in the standard errors. Third, the finding is not significantly driven by outliers. Moreover, in unreported tests, we replicate our estimates excluding major cities

Table 6. Random Implementation Test

Estimation	Kickstarter Campaigns	Self- Employment
μ of random β	0.00079	−0.00012
σ random β	0.01699	0.00696
Estimated β (rel time $t + 4$)	−0.084	−0.106
Replications	1,000	1,000
Z-score	−4.990907	−15.212486
p-value	$p < 0.001$	$p < 0.001$

(e.g., New York, San Francisco, Boston) to help rule out the possibility that our results are driven by an intensive response from a few large markets.¹⁵ Although there is a substantial concentration of Kickstarter campaigns in locations such as Los Angeles, New York, and San Francisco, results from these estimations suggest that, while the size of the effect does decrease when major metropolitan areas are excluded, they remain qualitatively similar.

The second major test, also a random implementation model, addresses concerns related to the concentration of city rollouts in early 2014. Following Greenwood and Agarwal (2016), we implement this test by swapping the implementation vectors across cities (i.e., New York receives the vector for Minneapolis, which receives the vector for Philadelphia, which receives the vector for Miami, and so on). We then replicate this process 500 times and store the coefficients. This test allows us to examine whether there are structural differences between cities that receive Uber and those that do not, which could have yielded a drop in Kickstarter campaigns even if Uber had not arrived. Results are in Table 7 and suggest that the average treatment effect is indistinguishable from zero once the vectors are swapped. Taken in sum, these tests suggest that neither outliers nor serial correlation are driving the results.

Other Ridesharing Services

Thus far, our results have focused on the effect of Uber X. However, other ridesharing services are in operation during the sample period. It is therefore incumbent upon us ensure that the omission of these other services in our primary estimation is not biasing our estimates. It is plausible, for example, that other *discount* ridesharing services may evoke the same theoretical tension as Uber X for the would-be entrepreneur. If this were the case, the would-be entrepreneur might begin driving for Lyft as opposed to Uber X. Moreover, *premium* ridesharing services, for example, Uber Black, are also in operation, providing an opportunity to evaluate our proposed theoretical mechanism, that is, necessity based entrepreneurship. The costs of participating as a driver for Uber Black (which requires a higher-end vehicle) are larger than those of

Table 7. Random Implementation Swapping City Vectors

DV:	Model 1 Kickstarter Campaigns	Model 2 Self- Employment
μ of ≥ 7 quarters until Uber X entry	−0.03488	−0.00124
σ of ≥ 7 quarters until Uber X entry	0.16173	0.03600
μ of 6 quarters until Uber X enters	−0.02482	0.00070
σ of 6 quarters until Uber X enters	0.10849	0.03557
μ of 5 quarters until Uber X enters	−0.01277	0.00079
σ of 5 quarters until Uber X enters	0.07529	0.02968
μ of 4 quarters until Uber X enters	−0.00552	−0.00072
σ of 4 quarters until Uber X enters	0.06108	0.03093
μ of 3 quarters until Uber X enters	−0.00347	0.00064
σ of 3 quarters until Uber X enters	0.04607	0.03440
μ of 2 quarters until Uber X enters	−0.00084	0.00007
σ of 2 quarters until Uber X enters	0.03276	0.02798
1 quarter until Uber X enters	Omitted	
μ of 0 quarters since Uber X entry	0.00417	−0.00065
σ of 0 quarters since Uber X entry	0.04225	0.02558
μ of 1 quarters since Uber X entry	0.00576	0.00079
σ of 1 quarters since Uber X entry	0.04340	0.03275
μ of 2 quarters since Uber X entry	0.00711	0.00270
σ of 2 quarters since Uber X entry	0.05115	0.03381
μ of 3 quarters since Uber X entry	0.00794	0.00213
σ of 3 quarters since Uber X entry	0.05506	0.03719
μ of 4 quarters since Uber X entry	0.00690	0.00157
σ of 4 quarters since Uber X entry	0.05712	0.04234
μ of 5 quarters since Uber X entry	0.00773	0.00495
σ of 5 quarters since Uber X entry	0.06214	0.04553
μ of 6 quarters since Uber X entry	0.01132	0.00813
σ of 6 quarters since Uber X entry	0.06989	0.05459
μ of ≥ 7 quarters since Uber X entry	0.01655	0.01046
σ of ≥ 7 quarters since Uber X entry	0.08327	0.06395
Replications	500	500
Number of EAnum	172	172
EA fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Time period	2012–2015	2012–2015

Uber X. As a result, if our proposed mechanism is correct, we should see a significantly smaller, if any, effect from Uber Black.

To address the effect of other discount ridesharing services, we first gathered the implementation schedules of Uber X's main competitors: Lyft, Sidecar, Flywheel, and Curb. Uber appears to have been the first to market in all but 15 economic activity areas. In 13 of the 15 EAs where Uber X was not the first to arrive, it was at most 2 months behind the other services in its arrival. In the remaining 2 EAs, it arrived 8 months later (Los Angeles, where Lyft's expansion was much of the impetus for Uber's nationwide rollout of Uber X) and 19 months later (Lincoln, NE, where Uber has recently launched). We then replicate Equation (1) with the definition of the relative time dummies adjusted such that they reflect the first implementation of any ridesharing service in a location (Uber X or a competitor). Results of this second analysis are presented in Models 1 and 2 of Table 8. Further, to explore any

Table 8. Consideration of Other Ridesharing Services

Platform:	Model 1 Uber X or competitor	Model 2 Uber X or competitor	Model 3 Uber Black	Model 4 Uber Black
DV:	<i>Campaigns</i>	<i>Self-Employed Persons</i>	<i>Campaigns</i>	<i>Self-Employed Persons</i>
≥7 quarters until platform entry	−0.0585 (0.115)	−0.0430 (0.0413)	−0.275 (0.193)	0.0182 (0.0708)
6 quarters until platform enters	0.0168 (0.0766)	−0.0315 (0.0287)	−0.125 (0.158)	−0.0151 (0.0684)
5 quarters until platform enters	−0.0480 (0.0562)	0.0241 (0.0239)	−0.0870 (0.0760)	−0.0522 (0.0483)
4 quarters until platform enters	−0.0173 (0.0350)	−0.0177 (0.0204)	−0.0256 (0.0530)	−0.0913* (0.0398)
3 quarters until platform enters	0.0128 (0.0331)	−0.0236 (0.0215)	−0.0270 (0.0522)	−0.0409 (0.0446)
2 quarters until platform enters	0.0108 (0.0199)	−0.0203 (0.0156)	−0.0828 (0.0522)	0.0170 (0.0403)
1 quarter until Uber X enters		Omitted		
0 quarters since platform entry	−0.0125 (0.0253)	−0.0312* (0.0159)	0.0733 (0.0558)	0.0103 (0.0307)
1 quarters since platform entry	0.00849 (0.0294)	−0.0661** (0.0220)	0.0801 (0.0504)	0.0376 (0.0305)
2 quarters since platform entry	−0.00400 (0.0309)	−0.0715** (0.0218)	0.0486 (0.0463)	0.00539 (0.0339)
3 quarters since platform entry	−0.0294 (0.0320)	−0.0702** (0.0258)	0.118* (0.0584)	−0.0161 (0.0342)
4 quarters since platform entry	−0.104* (0.0436)	−0.0702* (0.0313)	0.111+ (0.0619)	0.0512 (0.0496)
5 quarters since platform entry	−0.118** (0.0391)	−0.0914* (0.0368)	0.0801 (0.0650)	0.000268 (0.0483)
6 quarters since platform entry	−0.155*** (0.0448)	−0.0730+ (0.0442)	0.0391 (0.0664)	−0.0322 (0.0377)
≥7 quarters since platform entry	−0.240*** (0.0500)	−0.143** (0.0546)	−0.0255 (0.0799)	−0.0306 (0.0430)
Employment controls from Table 2	Yes	Yes	Yes	Yes
Observations	6,731	6,731	6,731	6,731
Number of EAnum	144	144	144	144
Model	Poisson	Poisson	Poisson	Poisson
EA fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes
Time period	2012–2015	2012–2015	2012–2015	2012–2015

Notes. Dependent variable *Campaigns* indicates the DV is the number of Kickstarter campaigns launched in *it*. Dependent variable *Self-Employed Persons* is the number self-employed individuals in *it*. Employment controls indicate the number of employed people, average weekly wage, total quarterly wages within the EA, percentage of unemployed people in the EA, and percentage of individuals in the EA with at least a bachelor's degree. Estimator is a PPML. Standard errors in parentheses are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.10$.

possible influence of *premium* ridesharing services, we gather data on the implementation schedule of Uber Black (again from the Uber blog). We then replicate the estimation of Equation (1) using a set of relative time dummies that reflect the rollout of Uber Black. These results are presented in Models 3 and 4 of Table 8.

The estimates add interesting nuance to our previous findings. First, we see in Models 1 and 2, under our adjusted definition of treatment (i.e., the entry

of *any* discount ridesharing service), that the results are unchanged. Second, we see in Models 3 and 4 that there is no significant effect of Uber Black entry on the volume of Kickstarter campaign launches or the level of CPS self-employment. Given that Uber Black does not present an easily accessible employment alternative for an average necessity-based entrepreneur (Uber Black relies heavily on professional livery drivers), this null result is consistent with our proposed

Table 9. Replication of Relative Time Model with EA-Specific Linear Trends

DV:	Model 1 <i>Campaigns</i>	Model 2 <i>Campaigns</i>	Model 3 <i>Self-Employed Persons</i>	Model 4 <i>Self-Employed Persons</i>
≥7 quarters until Uber X entry	−0.0155 (0.129)	−0.00836 (0.122)	−0.000975 (0.0407)	−0.0116 (0.0401)
6 quarters until Uber X enters	0.0398 (0.0757)	0.0486 (0.0715)	−0.00103 (0.0295)	−0.00380 (0.0300)
5 quarters until Uber X enters	−0.0261 (0.0572)	−0.0232 (0.0548)	0.0455+ (0.0235)	0.0450+ (0.0233)
4 quarters until Uber X enters	0.0114 (0.0396)	0.0175 (0.0378)	0.00124 (0.0236)	−0.00326 (0.0224)
3 quarters until Uber X enters	−0.00385 (0.0359)	0.00586 (0.0358)	0.0237 (0.0255)	0.0126 (0.0258)
2 quarters until Uber X enters	0.0307 (0.0217)	0.0366 (0.0233)	0.000713 (0.0156)	−0.00235 (0.0158)
1 quarter until Uber X enters			Omitted	
0 quarters since Uber X entry	−0.0315 (0.0236)	−0.0290 (0.0223)	−0.0214 (0.0174)	−0.0253 (0.0174)
1 quarters since Uber X entry	−0.0238 (0.0353)	−0.0140 (0.0299)	−0.0305 (0.0218)	−0.0437+ (0.0229)
2 quarters since Uber X entry	−0.0502 (0.0402)	−0.0449 (0.0332)	−0.0502* (0.0246)	−0.0557* (0.0256)
3 quarters since Uber X entry	−0.0904+ (0.0487)	−0.0901* (0.0383)	−0.0401 (0.0280)	−0.0467+ (0.0282)
4 quarters since Uber X entry	−0.147** (0.0530)	−0.147** (0.0464)	−0.0455 (0.0346)	−0.0592+ (0.0347)
5 quarters since Uber X entry	−0.153** (0.0564)	−0.144** (0.0440)	−0.0624+ (0.0360)	−0.0840* (0.0374)
6 quarters since Uber X entry	−0.193** (0.0682)	−0.187** (0.0590)	−0.0871+ (0.0478)	−0.0913+ (0.0474)
≥7 quarters since Uber X entry	−0.268*** (0.0528)	−0.258*** (0.0559)	−0.108* (0.0549)	−0.0792 (0.0530)
Employment controls from Table 2	No	Yes	No	Yes
Observations	6,731	6,731	6,731	6,731
Number of EAnum	144	144	144	144
Model	Poisson	Poisson	Poisson	Poisson
EA fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes
EA linear time trends	Yes	Yes	Yes	Yes
p-value of joint significance of time trends	0.00	0.00	0.00	0.00
Time period	2012–2015	2012–2015	2012–2015	2012–2015

Notes. Dependent variable *Campaigns* indicates the DV is the number of Kickstarter campaigns launched in *it*. Dependent variable *Self-Employed Persons* is the number self-employed individuals in *it*. Employment controls indicate the number of employed people, average weekly wage, total quarterly wages within the EA, percentage of unemployed people in the EA, and percentage of individuals in the EA with at least a bachelor's degree. Estimator is a PPML. Joint significance test is a likelihood ratio test comparing the model in this table with the similar model in Table 3. Standard errors in parentheses are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.10$.

mechanism, that is, a reduction in necessity-based entrepreneurship.

EA-Specific Time Trends

It is possible that idiosyncratic time trends exist across economic areas that our previous analyses have failed to account for. To resolve this concern, we replicate our estimations while including EA-specific linear time trends (in addition to EA fixed effects). The benefit of this model is that, although it sacrifices

power, it allows for varying trajectories in entrepreneurial activity across EAs. The results are presented in Table 9, where we see broad consistency with our main set of estimates. Likelihood ratio tests indicate that the EA-specific time trends provide a statistically significant increase in the fit of the models.

Empirical Extensions

We next consider analyses that (in tandem with our above results from Uber Black) help to better identify

the mechanism underlying the observed effect, that is, necessity-based entrepreneurship, which should result in a shift away from lower quality entrepreneurial projects and toward participation in the gig economy.

Failed vs. Successful Campaigns and Unincorporated vs. Incorporated Self-Employment

Our first concern relates to heterogeneity in the effects of gig-economy entry based on entrepreneur quality. To execute these tests, we proxy Kickstarter quality using fundraising outcomes, thereby allowing the market, that is, the crowd, to delineate between high- and low-quality ideas. We proxy self-employment quality capturing whether CPS respondents indicate that their business is unincorporated or incorporated, the key assumption being that incorporated businesses are more robust and higher quality since they are more likely to have employees and may have significantly larger costs to maintain.¹⁶ If the gig economy is reducing entrepreneurial activity among necessity-based entrepreneurs, who are generally of lower quality, we would anticipate the lion's share of the effect to accrue among crowdfunding campaigns with lesser fundraising success and among unincorporated entrepreneurs. To execute this test, we split Kickstarter campaigns into four buckets based on the percentage of target raised by the entrepreneur: (i) unfunded campaigns represents campaigns launched that received zero funding, (ii) partially funded campaigns represents campaigns that received some funding but did not reach their funding goal, (iii) funded campaigns represents campaigns that met their funding goal, and (iv) hyper-funded campaigns represents campaigns that received double their funding goal. We further split self-employment into two groups based on whether the CPS respondent indicates that the respondent's business is unincorporated or incorporated. We then replicate our estimations on each subsample. Results are in Table 10.

First, we examine Kickstarter outcomes. Consistent with the notion that gig-economy work shifts effort away from lower quality entrepreneurial projects, the largest share of the effect accrues to campaigns that receive no funding (Model 1). Further, we note in Models 2 and 3 that there is a moderate decrease in both partially funded campaigns and funded campaigns although the effect is intermittent. In Model 4, we observe no statistically significant change in the volume of hyper-funded campaigns (those that top 200% of their funding goal). Next, we compare unincorporated and incorporated CPS self-employment in Models 5 and 6. We see that the effect primarily occurs in unincorporated self-employment with little effect on incorporated self-employment. Taken in sum, these results lend credence to the idea that gig-economy platforms reduce entrepreneurial activity by allowing

individuals to pursue gig-economy employment rather than lower quality entrepreneurial projects (indicative of necessity-based entrepreneurship).

Total Dollars Pledged and Unemployment Rate

One further alternative explanation for the observed effects is that Uber X is choosing to enter relatively depressed economic areas. While our models include multiple controls to help account for this alternative explanation, we examine this possibility more deeply using two different dependent variables: total Kickstarter dollars pledged to local projects and the local unemployment rate. To the extent that capital is often provided by local individuals on crowdfunding platforms (Agrawal et al. 2015a), local funding totals on Kickstarter should provide a Kickstarter-specific proxy for local economic activity. The unemployment rate should also be a strong proxy for the health of the local economy.

If total dollars pledged remain stable after platform entry, it suggests that our results are driven by a shift in the campaigns that are launched locally following platform entry rather than the amount of capital that is available from the market. Results in Model 1 of Table 11 indicate no significant pretreatment or post-treatment differences in total dollars pledged by local funders on Kickstarter, suggesting that Uber X is not significantly correlated with the total capital supplied by funders (and thus merely associates with a reduction in the number of campaigns launched). Results in Model 2 of Table 11 indicate no significant effect on the local unemployment rate. These results argue against an alternative explanation whereby Uber strategically selects systematically downtrodden or growing locations for entry.

Examining Self-Reported Occupation

In our next empirical extension, we examine whether the entry of Uber X has a measurable effect on particular occupational categories. First, we examine whether Uber X reduces self-employment in occupations in which individuals would likely post on Kickstarter. This analysis helps close the gap between our two dependent variables and enables us to demonstrate that, in combination, they are capturing a cohesive phenomenon. As Mollick (2014) describes, most Kickstarter projects fall under the heading of technology (e.g., new computer programs) or art (e.g., photography). We thus identify self-employed individuals who list occupations that are likely to be engaged in technology- or art-oriented work (note that industry is not included in the CPS).¹⁷ In Table 11, Model 3, we examine how self-employment among these occupations changes following Uber X entry. Consistent with the Kickstarter results, we see a decline in self-employment in these occupational categories.

Table 10. Relative Time Model of the Effect of Uber X Entry on Underfunded vs. Overfunded Campaigns and Unincorporated vs. Incorporated Self-Employment

DV:	Model 1 <i>Unfunded Campaigns</i>	Model 2 <i>Partially Funded Campaigns</i>	Model 3 <i>Funded Campaigns</i>	Model 4 <i>Hyperfunded Campaigns</i>	Model 5 <i>Unincorporated Self-Employment</i>	Model 6 <i>Incorporated Self-Employment</i>
≥7 quarters until Uber X entry	−0.0332 (0.141)	0.166** (0.0572)	0.00927 (0.0664)	0.126 (0.118)	0.0124 (0.0415)	−0.0735 (0.0530)
6 quarters until Uber X enters	−0.00385 (0.142)	0.225** (0.0752)	0.0452 (0.0517)	0.0326 (0.108)	0.0332 (0.0355)	−0.0615 (0.0489)
5 quarters until Uber X enters	−0.190 (0.135)	−0.0873* (0.0495)	−0.0185 (0.0572)	0.104 (0.0763)	0.0466* (0.0281)	0.0461 (0.0385)
4 quarters until Uber X enters	−0.104 (0.0933)	−0.0158 (0.0457)	0.0229 (0.0412)	0.00844 (0.0694)	0.0143 (0.0265)	−0.0433 (0.0388)
3 quarters until Uber X enters	−0.0535 (0.0970)	−0.0118 (0.0530)	0.000271 (0.0314)	0.121 (0.0758)	0.0177 (0.0273)	−0.0382 (0.0407)
2 quarters until Uber X enters	0.0541 (0.0813)	0.0621* (0.0246)	−0.0207 (0.0334)	−0.0107 (0.0556)	0.00140 (0.0216)	−0.0137 (0.0291)
1 quarter until Uber X enters	Omitted					
0 quarters since Uber X entry	−0.0308 (0.0552)	0.0154 (0.0229)	−0.0606* (0.0263)	−0.0461 (0.0527)	−0.0246 (0.0176)	−0.0439 (0.0305)
1 quarters since Uber X entry	0.0145 (0.0639)	0.00838 (0.0309)	−0.0590* (0.0307)	0.0651 (0.0487)	−0.0405 (0.0283)	−0.0979** (0.0362)
2 quarters since Uber X entry	−0.0406 (0.0550)	−0.00375 (0.0358)	−0.0749* (0.0306)	−0.0563 (0.0555)	−0.0624* (0.0291)	−0.0499 (0.0415)
3 quarters since Uber X entry	−0.159* (0.0671)	−0.0530 (0.0453)	−0.0524 (0.0350)	−0.0471 (0.0613)	−0.0651* (0.0319)	−0.0198 (0.0414)
4 quarters since Uber X entry	−0.144* (0.0758)	−0.0750 (0.0462)	−0.1000** (0.0375)	0.0175 (0.0621)	−0.0808* (0.0318)	−0.0486 (0.0503)
5 quarters since Uber X entry	−0.208** (0.0743)	−0.0855* (0.0437)	−0.124** (0.0426)	−0.00284 (0.0736)	−0.112** (0.0369)	−0.0903 (0.0560)
6 quarters since Uber X entry	−0.162* (0.0860)	−0.0971* (0.0515)	−0.147** (0.0565)	−0.131 (0.0858)	−0.126* (0.0501)	−0.0319 (0.0638)
≥7 quarters since Uber X entry	−0.178* (0.0859)	−0.139** (0.0446)	−0.173** (0.0540)	−0.159 (0.0899)	−0.129* (0.0541)	−0.0260 (0.0780)
Employment controls from Table 2	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,731	6,731	6,714	6,473	6,731	6,731
Number of EAnum	144	144	143	138	144	144
Model	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
EA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time period	2012–2015	2012–2015	2012–2015	2012–2015	2012–2015	2012–2015

Notes. Dependent variable definitions related to the number of Kickstarter campaigns launched in *it*. *Unfunded Campaigns* indicates zero funding received. *Partially Funded Campaigns* indicates less than 100% funding received. *Funded Campaigns* indicates funding goal was met. *Hyperfunded Campaigns* indicates 200% of funding goal reached. Dependent variable *Unincorporated Self-Employment* is the number self-employed individuals who operate unincorporated businesses in *it*. Dependent variable *Incorporated Self-Employment* is the number self-employed individuals who operate incorporated firms in *it*. Employment controls indicate the number of employed people, average weekly wage, total quarterly wages within the EA, percentage of unemployed people in the EA, and percentage of individuals in the EA with at least a bachelor's degree. Estimator is a PPML. Standard errors in parentheses are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.10$.

The point estimate is slightly larger than what we see for the full set of occupational categories although it is also estimated less precisely as we might expect given that we are examining a sparser dependent variable. This pattern provides some evidence that the Kickstarter results and the CPS results are capturing the same phenomenon.¹⁸

Next, we examine whether Uber X has a measurable effect on local employment in the market for pro-

fessional drivers (e.g., taxi drivers, chauffeurs). Our argument is that individuals are working for Uber instead of engaging in lower quality entrepreneurial activity. However, we lack the individual-level panel data that would allow us to show this directly. It is therefore helpful to examine whether Uber's entry shifts local employment toward its industry, that is, driving. We once again draw upon the IPUMS-CPS data set, performing an analysis that is identical to that

Table 11. Relative Time Model of the Effect of Uber X Entry on Kickstarter Pledges, Unemployment Rate, and Self-Reporting as Driver in CPS

DV:	Model 1 <i>Dollars Pledged</i>	Model 2 <i>Unemployment Rate</i>	Model 3 <i>Self-Employment in Tech or Artistic Occupations</i>	Model 4 <i>Driver or Chauffeur as Primary Occupation</i>
≥7 quarters until Uber X entry	−0.0932 (0.205)	−0.00344 (0.00301)	0.0661 (0.0766)	−0.0863 (0.181)
6 quarters until Uber X enters	−0.00852 (0.347)	−0.00374 (0.00320)	0.105 (0.0796)	0.118 (0.189)
5 quarters until Uber X enters	0.207 (0.200)	−0.00368 (0.00242)	0.0893 (0.0791)	0.0110 (0.153)
4 quarters until Uber X enters	0.00245 (0.168)	0.000243 (0.00245)	0.0769 (0.0611)	−0.0740 (0.132)
3 quarters until Uber X enters	0.185 (0.126)	−0.000144 (0.00239)	0.0402 (0.0518)	−0.0844 (0.128)
2 quarters until Uber X enters	0.0962 (0.124)	−0.000738 (0.00211)	0.0329 (0.0619)	0.00157 (0.119)
1 quarter until Uber X enters			Omitted	
0 quarters since Uber X entry	0.0702 (0.208)	0.000105 (0.00168)	−0.0341 (0.0458)	0.153 (0.119)
1 quarters since Uber X entry	−0.00981 (0.138)	0.00161 (0.00252)	−0.0710 (0.0596)	0.174 (0.132)
2 quarters since Uber X entry	−0.0900 (0.169)	0.00280 (0.00236)	−0.000114 (0.0734)	0.123 (0.125)
3 quarters since Uber X entry	−0.176 (0.119)	0.00377 (0.00236)	−0.119 ⁺ (0.0716)	0.172 (0.133)
4 quarters since Uber X entry	−0.0744 (0.113)	0.000426 (0.00279)	−0.124 ⁺ (0.0750)	0.308 [*] (0.132)
5 quarters since Uber X entry	−0.150 (0.175)	0.00334 (0.00312)	−0.167 [*] (0.0821)	0.369 [*] (0.147)
6 quarters since Uber X entry	−0.0420 (0.132)	0.00404 (0.00328)	−0.131 (0.0950)	0.422 ^{**} (0.151)
≥7 quarters since Uber X entry	0.00357 (0.116)	0.00302 (0.00416)	−0.141 (0.121)	0.289 (0.185)
Employment controls from Table 2	Yes	Yes	Yes	Yes
Observations	6,731	6,731	6,618	6,041
R-squared		0.156		
Number of EAnum	144	144	141	128
Model	Poisson	OLS	Poisson	Poisson
EA fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes
Time period	2012–2015	2012–2015	2012–2015	2012–2015

Notes. Dependent variable *Dollars Pledged* is the number of dollars donated to Kickstarter campaigns in *i t*. *Unemployment Rate* is the percentage of individuals in the EA who are unemployed. Using year 2000 occupation codes, artistic occupations are defined as those beginning with 26, 27, 28, or 29, and tech occupations as those beginning with 10 or 11 (*Self-Employment in Tech or Artistic Occupations*). *Driver or Chauffeur as Primary Occupation* is the number of people self-reporting as drivers or chauffeurs in CPS-IPUMS. Employment controls indicate the number of employed people, average weekly wage, total quarterly wages within the EA, percentage of unemployed people in the EA, and percentage of individuals in the EA with at least a bachelor's degree. In Model 2, control for percentage of unemployment is replaced with control for percentage self-employed. Standard errors, in parentheses, are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.10$.

which we conduct for self-employment. Specifically, at the EA-month level, we regress the number of individuals who report their primary occupation as being a paid driver or chauffeur on the entry of Uber. We expect a positive and significant relationship between Uber X entry and individuals' self-reporting as a paid driver.

Results in Model 4 of Table 11 reveal the expected correlation. Notably, we observe an absence of pretreatment trends, indicating that, conditional on fixed effects and controls, Uber X does not appear to enter into cities where the taxi and limousine sectors are

already growing. Additionally, within one there is a 31% increase in individuals reporting driving as their primary occupation. It is useful to compare this effect size with those from self-employment, within which Uber X is associated with a 5% drop in self-employment one year after the introduction of the platform. While we cannot directly compare these elasticities, the fact that the increase in driver employment is qualitatively larger in magnitude than the decrease in self-employment gives us some confidence that our effect sizes are reasonable. Uber X will

undoubtedly draw workers away from activities other than self-employment. So, if a stronger effect on self-employment than on driver employment was observed, it would raise the concern that an unobserved third factor (other than Uber X) might be driving the downward trend in self-employment. Note that the number of observations in this analysis is smaller than the self-employment analysis because of a few EAs where no individuals report driver occupations during the period of study.

Hazard Model Predicting Uber X Entry

Finally, we examine whether measures of our dependent variables and control variables significantly influence the likelihood of Uber X's entry into an EA. In Table 12, we estimate a discrete time (logit) hazard model (Singer and Willett 1993). In this model, the unit

of observation is EA-month, and a 0/1 indicator of whether or not Uber X has entered the EA is the dependent variable. As is customary with survival models, we discard observations of an EA following Uber X entry. As predictors of entry, we include year and quarter fixed effects, Kickstarter campaign volume, number of self-employed persons, average weekly wage within the EA, total quarterly wages within the EA, the unemployment rate in the EA, the percentage of EA residents who have at least a bachelor's degree, and the number of employed people working in the EA.

Models 1–3 use the levels of these variables, and Models 4–6 use yearly changes (i.e., value of variable in month k of year t minus value of variable in month k of year $t - 1$). Looking at Models 1–3, the only statistically significant predictor is the percentage of EA

Table 12. Logit Hazard Model Predicting Uber X's Entry Into an Economic Area

DV:	Model 1 <i>Uber X Enters at Time t</i>	Model 2 <i>Uber X Enters at Time t</i>	Model 3 <i>Uber X Enters at Time t</i>	Model 4 <i>Uber X Enters at Time t</i>	Model 5 <i>Uber X Enters at Time t</i>	Model 6 <i>Uber X Enters at Time t</i>
<i>Campaigns</i>	0.00422 (0.00750)		0.00496 (0.00580)			
<i>Self-employment</i>		−5.61e−07 (4.26e−06)	−1.67e−06 (4.67e−06)			
<i>Employment</i>	−4.59e−07 (6.18e−07)	−2.97e−07 (6.52e−07)	−2.83e−07 (7.80e−07)			
<i>Avg. weekly wage</i>	0.00198 (0.00208)	0.00189 (0.00213)	0.00186 (0.00214)			
<i>Total quarterly wages</i>	8.45e−05 (5.84e−05)	8.36e−05 (5.36e−05)	8.33e−05 (5.97e−05)			
<i>Unemployment rate</i>	−0.386 (2.964)	−0.190 (2.965)	−0.325 (2.974)			
<i>% with bach. degree</i>	1.886* (0.817)	1.920* (0.790)	1.879* (0.806)			
<i>Campaigns (yearly Δ)</i>				−1.19e−05 (1.34e−05)		−1.23e−05 (1.37e−05)
<i>Self-employment (yearly Δ)</i>					4.34e−09 (6.50e−09)	4.59e−09 (6.34e−09)
<i>Employment (yearly Δ)</i>				3.01e−09+ (1.61e−09)	2.76e−09 (2.06e−09)	2.55e−09 (1.87e−09)
<i>Avg. weekly wage (yearly Δ)</i>				2.83e−06 (1.81e−06)	3.01e−06+ (1.71e−06)	2.73e−06 (1.80e−06)
<i>Tot. quarterly wages (yearly Δ)</i>				−1.54e−07** (4.75e−08)	−1.48e−07** (4.82e−08)	−1.53e−07** (5.00e−08)
<i>Unemploy. rate (yearly Δ)</i>				−0.000202 (0.000634)	−0.000224 (0.000650)	−0.000272 (0.000638)
<i>% with bach. degree (yearly Δ)</i>				−5.72e−05 (0.000408)	−1.11e−05 (0.000425)	−9.18e−05 (0.000419)
Observations	5,456	5,456	5,456	5,420	5,420	5,420
Model	Logit	Logit	Logit	Logit	Logit	Logit
Year and quarterly fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time period	2012–2015	2012–2015	2012–2015	2012–2015	2012–2015	2012–2015

Notes. Dependent variable *Uber X Enters at Time t* is a dummy that takes the value of 1 in the period in which Uber X first enters the economic area. EA exits the sample in period following Uber X entry. Yearly change variables indicate the difference between the value of the explanatory variable in month t as compared with the value of the explanatory variable in the same month of the previous year. Standard errors, in parentheses, are robust and clustered by EA.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.10$.

residents with at least a bachelor's degree, which has a positive relationship with Uber X entry. This relationship makes sense because college-educated individuals are more likely to use ridesharing services (Rayle et al. 2016). In Models 4–6, only change in total wages, which has a negative relationship with Uber X entry, is significant. This could be explained by Uber X entering areas with declining wages and thus an ample supply of individuals who might be interested in driving for Uber X. Critically, we see that the level and yearly change in our measures of entrepreneurial activity never significantly affect Uber entry.

Ridesharing Driver Survey

Our archival results tell a compelling story. Nonetheless, to better establish the face validity of our findings, we undertook a direct survey of Uber and Lyft drivers. We recruited a sample of 179 respondents via online discussion forums that host Uber and Lyft driver communities. We obtained 20 responses by sharing a link to our survey on the r/uberdrivers and r/lyftdrivers subreddits on Reddit.com. The remaining 159 responses came via a data-collection agreement with UberPeople.net, wherein we paid the forum host on a cost-per-click (CPC) basis of \$1 per click up to \$300. As compensation for completing the survey, respondents entered a lottery for one of five \$50 Amazon Gift Cards.

In the survey, we collected background information on drivers' experience in ridesharing and asked drivers to indicate (i) whether they viewed themselves as entrepreneurs, (ii) whether they had ever considered starting a small business, (iii) whether they currently operate a small business in parallel to their work as a driver, and (iv) how they would go about raising money for a new small business. Responses largely aligned with the findings of our archival analyses (the full survey is in the appendix).

First, results indicate that gig-economy employment is inherently different from traditional entrepreneurship. Whereas 14 respondents (7.78%) view themselves as entrepreneurs, nearly an order of magnitude more, that is, 135 or 75%, view themselves as independent contractors. Second, our results indicate that a large proportion of individuals had an interest in launching a small business before they became employed in the gig economy. Specifically, 122 (67.78%) respondents indicate that they considered launching a small business, yet more than two thirds of these individuals (86 respondents, or 48% of our overall sample) had not ultimately done so. Finally, survey results indicate that participants in the gig economy are not only aware of crowdfunding as an option for sourcing capital to support a new venture, but many have direct experience in executing a crowdfunding campaign. Strikingly, 38 respondents (21.11%) indicate that they would

employ crowdfunding to raise money for a new business, and 12 (6.67%) report they had prior experience organizing a crowdfunding campaign.

These survey proportions align well with our archival estimates. Consider that Uber hosted approximately 327,000 (Carson 2015) drivers in the United States at the end of our study period. The survey suggests that ~157,000 (48%) would have considered starting a small business yet chose not to do so. Survey results further suggest that, were those individuals to have started a small business, ~33,000 (21%) would have raised initial capital via crowdfunding. Kickstarter has hosted approximately 320,000 projects over seven years across the country, or ~45,000 per year, on average; thus, our archival estimate of a 14% annual decline translates to a loss of 7,300 campaigns following Uber's entry (note: $[45,000/(1 - 0.14)] - 45,000 = 7,325$) if Uber were to have entered every location. Notably, however, Uber did not enter every location; thus, the number of "lost" campaigns is perhaps closer to ~5,000 per year. This estimate is quite feasible if we assume that the 33,000 drivers our survey implies would have launched a business in the absence of Uber and who would have employed crowdfunding to do so were distributed over the six years between Uber's founding in 2009 and the end of our study window in 2015.

Discussion

An interesting tension arises when one considers how the entry of gig-economy platforms might influence entrepreneurial activity. On the one hand, gig-economy employment might provide individuals with flexibility and resources that enable them to engage in the creation of a new venture, thus increasing total entrepreneurial activity. On the other, the presence of gig-economy platforms may provide a desirable employment alternative for would-be necessity-based entrepreneurs, thus reducing total entrepreneurial activity. Exploiting a multi-treatment, difference-in-difference specification around the entry of gig-economy platform Uber X, we find consistent evidence of a negative effect of gig-economy platform entry on entrepreneurial activity as measured by the volume of crowdfunding campaign launches and individuals reporting self-employment in the U.S. Census Bureau's Current Population Survey. Further, we observe that the effect of entry is stronger when the entering platform bears lower fixed cost of participation (e.g., comparing Uber X and Uber Black), and the effect manifests primarily as a reduction in unsuccessful crowdfunding campaigns and unincorporated self-employment, that is, lower quality entrepreneurs. These findings suggest that gig-economy platforms provide necessity-based entrepreneurs with a preferable, alternative source of employment.

This work makes four substantive contributions, which offer exciting opportunities for future research. First, our results provide a glimpse into the supply side of gig-economy markets, suggesting that individuals who take up jobs on these platforms may be directing their efforts away from underperforming entrepreneurial activities. While researchers have considered the effect of platform entry on the demand side of markets (e.g., on competitors (Seamans and Zhu 2013, Zervas et al. 2015) or consumers (Edelman and Luca 2014, Greenwood and Wattal 2017)), research regarding the supply side of these markets remains nascent. We point toward the importance of investigating the supply side of these platforms and illustrate how their creation is influencing the labor market by creating new employment options for individuals with, potentially, a limited set of alternatives.

While we find that, on average, gig-economy platforms reduce entrepreneurial activity, an interesting opportunity for future work lies in identifying the conditions under which these platforms do supply some would-be entrepreneurs with the slack resources they require to pursue their ventures. The fact that numerous respondents in our survey indicated that they operate a small business in parallel to driving for a ridesharing service underscores the potential for this line of work. Researchers might explore specific geographies or business domains in which firm founding is critically dependent upon the availability of slack resources. The new Contingent Work Supplement to the CPS (Kopp and Edgar 2017) may provide researchers with the micro data necessary to identify the individuals who are able to get entrepreneurial projects off the ground as a result of the flexibility provided by gig employment.

Our second contribution lies with our use of a novel measure for entrepreneurial activity: the rate and volume of crowdfunding campaign launches. Crowdfunding data sees increasing adoption among entrepreneurship and innovation scholars (Agrawal et al. 2015a, Mollick 2014), and our work adds to this literature in multiple ways. For example, we demonstrate the value of using the market's response to a crowdfunding initiative as a proxy for the quality of the entrepreneurial idea. In doing so, we also point toward the value of examining both successful and failed campaigns as well as the possibility of using crowdfunding data to examine failed entrepreneurship. In addition, we provide evidence for the use of crowdfunding as a measure of entrepreneurship (Mollick and Kuppaswamy 2014) by documenting its consistency with more traditional measures, such as reports of self-employment on the U.S. Census Bureau's CPS. We hope that future work continues down this path, which has recently been eased by the

public availability of project-level data from the Haas-Berkeley Crowdfunding website.

A third contribution applies to the crowdfunding literature. Uber X's differential effect on more versus less successful campaigns suggests a positive spillover for crowdfunding marketplaces. The entry of gig-economy platforms appears to help separate wheat from chaff and reduce "noise" in such marketplaces by reducing the number of less successful campaigns. Such declines in low-quality campaigns may reduce the cognitive burden and search costs imposed on crowd-financers, thereby mitigating adverse selection and helping to ensure the sustainability of the market. With a drop in low-quality campaigns, the crowd can shift its focus and wealth to higher quality campaigns and, in theory, achieve greater returns or make a more substantive impact. This contribution points toward the value of studying "noise" on these platforms more closely. What other types of quasi-exogenous events affect the posting of lower quality or higher quality projects (Geva et al. 2016) remains an open question as does the efficacy of various platform features at filtering out lower quality campaigns and the trade-offs platforms incur from leveraging such features.

Finally, this work offers insights for policy makers who are currently debating the legality of services such as Uber as well as the growing literature on the broader societal impacts of information systems (Bapna et al. 2016, Chan and Ghose 2014, Chan et al. 2016, Parker et al. 2016, Rhue 2015). Although our results do not capture potential reductions in employment within disrupted industries (e.g., taxis (Cramer and Krueger 2016), hotels (Zervas et al. 2015), or newspapers (Seamans and Zhu 2013)), they do suggest that these platforms may provide options for the unemployed and underemployed in other sectors. Policy makers may find it valuable to consider these potential effects when deciding whether to accommodate these platforms.

From a welfare perspective, it is difficult to make overarching claims about the long-term effects that result from the patterns we document in this study. To the extent that Uber X primarily reduces entrepreneurial activity that would likely have ended in failure, it is tempting to conclude that the patterns we document are mostly welfare enhancing. However, it is possible that some of the entrepreneurial ideas that are not pursued because of the availability of a gig-economy job might have otherwise blossomed into viable businesses that would have created jobs and led to useful innovations. Were some of these unexpected successes to have led to blockbuster products, for example, it is possible that any welfare losses would be cancelled out. As it is difficult to tease apart these competing effects with the short panel of data that we have, future work is needed to understand the full, long-term

implications of shifting labor away from lower quality entrepreneurship and toward gig-economy employment. This question goes to the heart of a broader debate among researchers and policy makers about whether or not the gig economy is welfare enhancing, particularly for workers who supply labor to the platforms (Heller 2017).

This work is subject to a number of limitations, which also offer rich opportunities for future research. First, as discussed, our analysis is dependent on the key assumption that the entry and timing of gig-economy platforms is exogenous with respect to entrepreneurial activity, conditional upon controls. While we have taken numerous steps to evaluate the validity of this assumption (e.g., examining pretreatment trends across numerous models), it is important to emphasize that the entry of Uber X into particular cities is not random, and future work that is able to identify valid instruments or conduct field experiments will make an important contribution.

Second, we lack individual-level panel data that would allow us to track specific people over time and observe them transitioning into employment with Uber X. This limitation forces us to use EA-level data, which, in turn, creates a loose coupling between our treatment and our ultimate outcomes because we are unable to tell whether the same individuals who drive for Uber are the same individuals who are not engaging in entrepreneurial activity. This also prevents us from understanding *for whom* the availability of gig-economy work might discourage or enable entrepreneurship. Future researchers with access to such detailed data could make an important impact because they would be able to more directly study the micro foundations of gig-economy participation.

Finally, it remains possible that the scheduling flexibility afforded by employment in these marketplaces may enable individuals to pursue other, nonentrepreneurial activities that are also subject to scheduling constraints or require other forms of slack resources, for example, pursuing an education, being a caregiver, or job seeking following a layoff. This observation suggests that there may be rich opportunities for future work exploring the gig economy's role in various nonentrepreneurial activities. Such work would help us to better understand why individuals supply labor to these platforms. More generally, it is our hope that this work presents an initial step in a broader and deeper examination of the trade-offs and choices that individuals make when participating in the gig economy, an examination that we hope will provide insights to policy makers, managers, and potential participants in the gig economy.

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Appendix

What ridesharing service do you drive for most often?

- ☐ Uber
- ☐ Lyft
- ☐ Other

How do you view your relationship with the ridesharing platform (e.g., Uber, Lyft) for which you drive?

- ☐ I am an employee of the platform.
- ☐ I am an entrepreneur.
- ☐ I am an independent contractor.
- ☐ I am not sure.

Approximately how long have you been driving for a ridesharing provider (e.g., Uber, Lyft)?

_____ Years

Approximately how many hours do you drive for ride sharing services in an average week?

_____ Hours/Week

Have you ever considered starting your own small business (aside from driving for a ridesharing service)?

- ☐ Yes
- ☐ No

Do you currently own and operate a small business (aside from driving for a ridesharing service)?

- ☐ Yes
- ☐ No

If you were to start your own small business, how would you raise initial funding? Select all answers that apply.

- ☐ Personal savings
- ☐ Credit card
- ☐ Bank loan
- ☐ Borrow from friends/family
- ☐ Crowdfunding (e.g., Kickstarter)
- ☐ Other

Have you ever organized a crowdfunding campaign?

- ☐ Yes
- ☐ No

Endnotes

¹ Scholars have recently debated the extent to which gig economy work differs from traditional temporary work arrangements. The primary distinction emphasized by most observers, including policy makers (e.g., Telles 2016), is the digital component of such firms. Such

components significantly lower the transaction costs of matching temporary workers to customers and monitoring service quality, thereby enabling a larger scale than previously possible. Emblematic of this idea is the decision of the Census Bureau to reintroduce the Contingent Worker Supplement to the CPS in 2017 after a decades-long hiatus. One proposed question for this survey highlights the changing nature of temporary work: “Some workers find short, in-person jobs or tasks through companies that connect them directly with customers using a website or mobile app. These companies also coordinate payment for the service through the app. Does this describe any work you did last week?” (Kopp and Edgar 2017, p. 12).

²When referring to “entrepreneurial activity,” we exclude employment on the gig-economy platform itself for a number of reasons. As an example, recent class action lawsuits filed against Uber (e.g., <http://uberlawsuit.com/>, <http://www.uberlitigation.com/>) indicate that many service providers view themselves as employees of the firm rather than independent contractors. Moreover, we completed a small-scale survey of service providers, the overwhelming majority of whom reported that they did not view themselves as entrepreneurs.

³This calculation is based on an observed median campaign request amount of \$5,200, 1,440 EA-quarter observations in which Uber X was active, and a conservative estimate of Uber X’s average marginal effect (−0.010) on the logged count of campaign launches. This calculation was performed as follows: $\exp(0.010)$ campaigns per period * 1,440 periods * \$5,200 = \$7,563,255.65.

⁴<http://syntaxi.net/2013/03/24/let-s-explore-kickstarter-s-api/> (last accessed February 19, 2018).

⁵<https://cps.ipums.org/cps/> (last accessed February 19, 2018).

⁶Based on the survey we report at the end of the paper, most Uber drivers do not consider themselves to be self-employed, so it appears likely that these individuals would not refer to themselves as such when responding to the CPS.

⁷<http://afrf.hrsa.gov/> (last accessed February 19, 2018).

⁸In robustness checks, we explore whether CPS effects manifest inside technical and artistic occupations, which are likely to be involved in Kickstarter campaigns. We do this to ensure the dependent variables are capturing parts of the same phenomenon.

⁹An explicit examination of other ridesharing services (e.g., Lyft, Sidecar, Uber Black) is reported in Table 8.

¹⁰Replication of our estimations using Uber Black yields no significant correlation.

¹¹A formal test of the skewness on the entry of Uber X into local areas is insignificant ($p = 0.155$), suggesting that the implementations are sufficiently distributed over time. A QQ plot of the normality of entry times corroborates this test. We thank the anonymous reviewer for suggesting these tests.

¹²An interesting question is whether these results are driven by the intensive versus extensive margins. To explore this more directly in the Kickstarter data, we estimate conditional logit models in which the dependent variable takes a value of one for various low thresholds of Kickstarter activity (e.g., posting of >0 projects in an EA-month). The treatment effects are not significant in any of these models (results available from the authors), suggesting that results manifest primarily along the intensive margin. We thank the anonymous reviewer for recommending this analysis.

¹³We thank the anonymous reviewer for suggesting this analysis.

¹⁴We opt for this approach over the alternative suggestion by Bertrand et al. (2004) to collapse panels into pairs of pre and post observations because our treatments are staggered in time.

¹⁵We thank the anonymous reviewer for this suggestion.

¹⁶This is the best proxy for self-employment quality that is available in the monthly CPS files. Other measures, such as earnings from self-employment, size of firm, etc., are not provided.

¹⁷Art-oriented occupations include arts, design, entertainment, sports, and media occupations (2% of employment; 5.7% of self-employment in the CPS data we analyze). Technology-related occupations include occupation codes beginning with 10 or 11 (i.e., computer-related occupations; 4% of employment, 4% of self-employment). We rely on the 2000 OCC codes; see <https://usa.ipums.org/usa/volii/occ2000.shtml> for the list of occupation categories.

¹⁸We thank the anonymous reviewer for suggesting this analysis.

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